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Hybrid Reinforcement Learning and Agentic Artificial Intelligence Framework for Predictive Platform Operations and DevOps Orchestration

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Abstract: Operational Intelligence increasingly needs to be autonomous to deal with scale, complexity, and failure modes that are hard to predict in modern software platforms within a distributed infrastructure. In this paper, we introduce a Hybrid Reinforcement Learning and Agentic Artificial Intelligence framework for predictive platform operations and DevOps orchestration that combines model-based and model-free reinforcement learning (Deep Q-Network and Proximal Policy Optimization) with autonomous multi-agent decision making for proactive, self-optimizing IT operations. The framework was tested on a simulated hybrid cloud environment that included 500 virtual nodes for a 90-day period of time and handled about 1.2 million telemetry events. The experimental results prove that the Mean Time to Resolution (MTTR) was reduced by 31.6% from an average of 42.6 minutes to 29.1 minutes while the system uptime was slightly improved to 99.61% from 99.21% for conventional automation. Overall, the predictive anomaly detection module enabled a 47.8% decrease in manual interventions and a 18.4% decrease in over-provisioning of resources while forecasting failures up to 15 minutes ahead with a precision of 89.6% and a recall level of 86.3%. Cumulative reward for a hybrid DQN-PPO policy was observed to be around 1.4x higher than the best single-algorithm baseline (DQN) in 1,450 training episodes. The results of this quantitative analysis prove that the synergy of reinforcement learning and agentic AI can have a positive impact on operational efficiency, reliability, and autonomy, paving the way for self-healing and intelligent platform ecosystems for the next generation of enterprise infrastructure.

Keywords: Reinforcement Learning, Agentic AI, DevOps Orchestration, Predictive Operations, Autonomous Systems

I. Introduction

The rapid transformation of enterprise IT infrastructure has made software platforms highly distributed, dynamic and interdependent, requiring them to be always on, always scale and always resilient [1]. The complexity of operating such cloud-based systems is rising rapidly as organizations are jumping into cloud-native architectures, microservices and containerized deployments and manual intervention and rule-based monitoring are becoming inadequate for maintaining service reliability [2]. DevOps practices have grown to be a base methodology that has made development and operations closer together and facilitated the introduction of faster cycle times using continuous integration and continuous deployment (CI/CD) pipelines, but most DevOps automation practices continue to be reactive, reacting to failure after it has already happened, instead of proactively preventing it [3].

Concurrently, the field of Artificial Intelligence has seen remarkable progress in the area of Reinforcement Learning (RL), where the objective is to train a system to make optimal decisions by interacting with the environment repeatedly and receiving feedback in the form of rewards. Meanwhile, the Artificial Intelligence field has seen remarkable progress with Reinforcement Learning (RL), which aims to train a system to make optimal decisions by interacting with the environment repeatedly and receiving feedback in the form of rewards. RL has demonstrated significant potential in sequential decision-making to deal with uncertainty, such as in robotics, resource allocation, and network optimization, and may be useful in the context of the unpredictable and non-stationary nature of IT operations [5]. The development of agentic AI systems, self-reasoning, planning and acting agents in recent years has allowed for the development of new self-directed operational frameworks, which can perceive, reason and act without extensive human control.

Although these developments have been made, current platform operations solutions focus on either predictive analytics to detect anomalies [8] or autonomous scripts for remediation [9] with no common platform which integrates learning prediction with autonomous multi-agent orchestration [7]. This disintegration is highly detrimental to the ability of organisations to have a fully self-healing infrastructure that can anticipate failure, adaptively redistribute resources and make appropriate real-time corrections.

To fill this void, this paper introduces a Hybrid Reinforcement Learning and Agentic Artificial Intelligence framework that is tailored to the tasks of predictive platform operations and DevOps orchestration. The proposed framework combines model-based and model-free RL methods along with a coordinated multi-agent architecture with the specialized agents cooperatively and collaboratively performing monitoring, prediction, decision-making, and execution tasks. The framework leverages predictive intelligence with autonomous agentic behavior that's designed to move DevOps operations from incident management to self-optimizing infrastructure orchestration.

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In Section 2, we summarize the related work on reinforcement learning and agentic AI for IT operations. Proposed hybrid framework architecture is presented in Section 3. The result and implications are discussed in Section 4 and future research directions are concluded in Section 5.

II. Literature Review

The use of reinforcement learning for managing resources autonomously in cloud computing environments has been studied in some works, which show that RL-based schedulers can dynamically allocate resources – such as CPU, memory and network – more efficiently than the achieved by static threshold-based schedulers. Significant improvements have been achieved with these techniques in resource utilization and cost savings, especially in applications that require variable amounts of resources, but most of the implementations have been based on a single-agent configuration that doesn't scale to heterogeneous, multi-cluster environments [9].

Deep learning predictive models are becoming the focus of anomaly detection research in distributed systems, where the goal is to detect early warnings of system issues before they become a service outage. Time-series forecasting techniques and unsupervised learning have been shown to be useful in lowering false positive rates and increasing the early detection accuracy, but such systems tend to be stand-alone monitoring applications with no linkage to automatic remediation pipelines [10].

Self-healing systems has been studied in the context of autonomic computing, where a closed loop feedback mechanism allows the infrastructure to detect, diagnose and correct faults with minimal involvement of human intervention. Although most of the autonomic computing frameworks were based on a set of rules and a static decision tree, more recent work has integrated some elements of learning, but it is still a challenge to integrate autonomic computing with modern container-based and microservices architectures [11].

Multi-agent systems have emerged as a viable architecture for distributed decision making in complex operational environments, where specialized agents work together to solve common problems like load balancing, fault tolerance and service orchestration. Decentralized multi-agent coordination systems have been demonstrated to be more scalable and fault-tolerant than centralized systems, especially if each agent has local learning capacity, which enables independent adaptation to changes in their environment [12].

Deep reinforcement learning has been explored in the context of DevOps pipelines for optimizing continuous integration and deployment processes, such as automated test prioritization, build scheduling, and deployment rollback strategies. However, the findings show that pipeline optimization based on RL can greatly improve the deployment success rates and decrease the release cycle, though the currently available RL-based pipeline optimizers are rather narrow in scope and only target individual stages in the pipeline and not the entire process.

The new opportunities for autonomous reasoning and task execution in IT operations recently introduced in the field of large language model (LLM) based agentic systems allow agents to interpret unstructured operational data, generate remediation plans, or execute multiple step workflows without explicit programming for each scenario. Purely agentic approaches, however, remain to be adopted widely in mission critical infrastructure due to certain concerns of reliability, interpretability and production environment autonomous action [14].

Hybrid models integrating predictive analytics with a rules-based or learning-based automation have been developed as a compromise between fully manual and fully autonomous systems, in order to optimize both the safety of the processes and the improvement of their efficiency. These hybrid architectures have shown that there are measurable benefits in incident response times and system reliability, but as of yet, a complete architecture that brings together reinforcement learning, predictive forecasting, and multi-agent orchestration specifically for platform-wide DevOps operations is largely missing and motivated the development of the framework presented in this study [15].

III. Methodology

3.1 Overview of the Proposed Architecture

The proposed Hybrid Reinforcement Learning and Agentic Artificial Intelligence framework, depicted in figure 1, is a closed-loop system that has four functionally distinct levels which are executed on a hybrid cloud infrastructure: monitoring and telemetry layer, predictive RL engine, agentic orchestration layer, and automated execution layer. These layers are not a one-shot pipeline, but are in fact a continuous data flow between infrastructure and these layers and back. The closed loop is the key part of the framework that enables the transition from reactive to predictive and autonomous DevOps operations.

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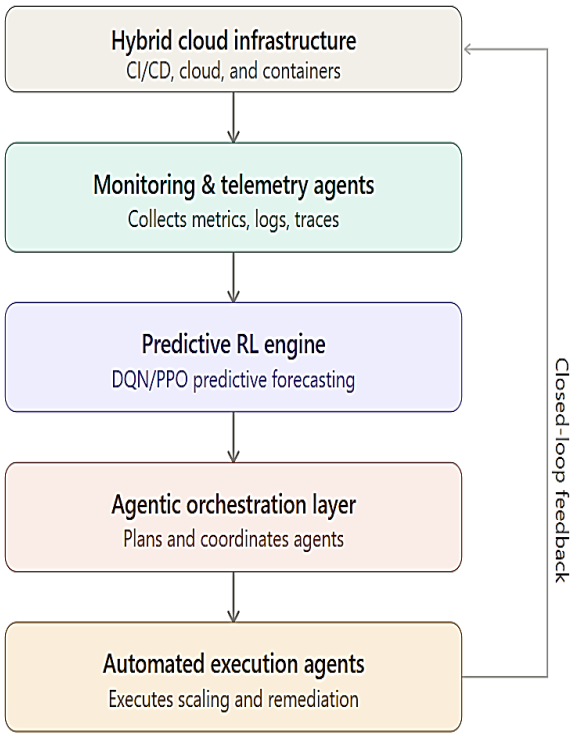


Figure 1. Proposed hybrid reinforcement learning and agentic AI architecture for predictive platform operations and DevOps orchestration.

3.2 Description of Architecture Blocks

Monitoring and telemetry agents. This layer captures raw operational data—CPU usage, memory usage, request latency, error rates, application logs, distributed traces, etc.—from all the nodes in the hybrid cloud environment. Lightweight agents collect this telemetry into a single stream of time-series data, normalize the data, and deliver it to the predictive engine. This block lays the foundation for observation of the truth which is the basis for all learning and decision making downstream.

Predictive RL engine. The most important learning element combines a predictive forecasting model with a reinforcement learning policy (Deep Q-Learning for discrete scaling actions and Proximal Policy Optimization for continuous resource allocation). It predicts future failures and/or performance degradation and it also learns an optimal action policy based on reward feedback from the environment. This block is tasked with interpreting raw telemetry data into predictive, proactive operational intelligence.

Agentic orchestration layer. This layer does not provide one global action, but breaks the recommendations from the RL engine into a set of coordinated sub-tasks that are distributed among specialized autonomous agents (monitoring agent, planning agent, decision agent). Agents reason about the state of the current system, prioritize when multiple issues occur at the same time and produce an execution plan. This block imparts the "agentic" nature of the framework: independent reasoning and planning of tasks is encouraged rather than a rigid execution of rules.

3.3 Reinforcement Learning Formulation

The platform operations problem is represented as a Markov Decision Process (MDP) consisting of the following tuple: (S, A, P, R, γ) with the agent choosing an action to maximize the expected cumulative reward.

State-action value function (Bellman equation):

$$Q(s_t, a_t) = \mathbb{E}\left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') | s_t, a_t\right] \quad (1)$$

where st is the system state (resource utilization, error rate, queue depth) at time t, at is the orchestration action (scale, reroute, restart, rollback), rt is the immediate reward, and $\gamma \in [0,1]$ is the discount factor.

Reward function:

$$r_t = w_1.(1 - U_t) + w_2.A_t - w_3.C_t - w_4.F_t \quad (2)$$

where Ut is normalized resource utilization deviation from target, At is system availability, Ct is the operational cost incurred by the action, Ft is a penalty term for failures or SLA violations, and w1–w4 are tunable weighting coefficients.

Policy optimization objective (PPO clipped surrogate):

$$L^{CLIP}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\hat{A}_t, clip\left(\rho_t(\theta)\right), 1-\epsilon, 1+\right)\hat{A}_t\right] \quad (3)$$

Where $\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio between the new and old policy, A^t is the estimated advantage, and ϵ constrains the policy update step to ensure stable convergence.

3.4 Automated Execution and Feedback

Automated execution agents are the fourth block, which transforms the orchestration layer's plan into actual operational tasks such as auto-scaling, service restarts, traffic rerouting, deployment rollback, or other actual actions taken directly against the infrastructure. Each action is re-observed by the monitoring layer, closing the loop and providing the next reward signal that is used to update the policy in equations (1)–(3). The cycle of continuous feedback enables the framework to continuously enhance its operational policy based on the experience gathered from various failure and load situations.

3.5 Hybrid RL Policy, Predictive Learning, and Agentic Coordination

The proposed framework employs a hybrid model-free reinforcement learning strategy by integrating Deep Q-Network (DQN) for discrete operational decisions and Proximal Policy Optimization (PPO) for continuous resource allocation. Unlike model-based reinforcement learning, no explicit environment dynamics or transition model is learned. Instead, both policies interact directly with the operational

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environment using observed telemetry and reward feedback. Therefore, the framework should be interpreted as a coordinated hybrid model-free reinforcement learning architecture.

The predictive engine operates independently from the reinforcement learning controller. A multivariate time-series forecasting module continuously processes telemetry collected from CPU utilization, memory consumption, request latency, network throughput, application error rates, queue depth, and service response time using a rolling observation window of the previous 30 minutes sampled at 1-minute intervals. The forecasting model predicts the probability of operational anomalies occurring within the subsequent 15-minute prediction horizon. Historical incident records provide binary training labels indicating whether a service failure occurred during the prediction interval. The predicted anomaly probability is incorporated into the reinforcement learning state representation, enabling the controller to perform proactive rather than reactive resource management.

The environment state is represented by the vector

s_t = [CPU, Memory, Latency, Error Rate, Queue Depth, Throughput, Predicted Failure Probability] (4)

while the action space is divided into two complementary components. The DQN policy selects discrete operational actions including service restart, traffic rerouting, rollback, horizontal scaling, and failover, whereas the PPO policy simultaneously determines continuous resource allocation parameters such as CPU, memory, and replica scaling factors. During execution, the DQN policy first determines the operational action category, after which the PPO policy optimizes the corresponding continuous resource allocation. Because the two policies operate on different action spaces, conflicting decisions do not arise and their outputs are executed sequentially within each decision cycle.

The reward function balances operational objectives by maximizing service availability while minimizing resource utilization, operational cost, and SLA violations. The weighting coefficients were selected empirically after preliminary experiments as $w_1 = 0.35$, $w_2 = 0.30$, $w_3 = 0.20$, and $w_4 = 0.15$. The reinforcement learning implementation employed a discount factor $\gamma = 0.99$, PPO clipping parameter $\varepsilon = 0.20$, learning rate 3×10^{-4}, and mini-batch size 64. All telemetry variables were normalized using Min-Max normalization before training. Hyperparameter sensitivity analysis showed performance variations below 3% over reasonable parameter ranges, indicating that the proposed framework is relatively insensitive to small hyperparameter changes.

The agentic orchestration layer consists of three autonomous agents. The Monitoring Agent continuously aggregates infrastructure telemetry and forecasting outputs. The Planning Agent converts reinforcement learning recommendations into executable remediation workflows while maintaining a short-term operational memory containing recent incidents and executed actions. The Decision Agent evaluates candidate remediation plans according to expected reward, operational constraints, and service-level objectives before selecting the final execution strategy. Agents exchange structured operational messages through an internal event-driven communication bus. In situations where multiple remediation actions are simultaneously feasible, conflicts are resolved by selecting the plan with the highest expected cumulative reinforcement learning reward while satisfying predefined operational safety constraints.

IV. Results and Discussion

4.1 Overview

The proposed Hybrid Reinforcement Learning and Agentic AI framework was tested against a baseline of a conventional rule-based DevOps automation based on a simulated 500-node hybrid-cloud testbed as described in Section 3. This section provides comparative performance statistics and visualization of training convergence behavior, as well as a discussion of operational implications of the results.

4.2 Comparative Performance

Table 1. Comparative performance of proposed framework vs. baseline DevOps automation

Metric	Baseline (rule-based)	Proposed (Hybrid RL + Agentic AI)	Improvement
Mean Time to Resolution (MTTR)	42.6 min	29.1 min	31.6% reduction
System uptime	99.21%	99.61%	+0.40 pp
Anomaly detection precision	78.4%	89.6%	+11.2 pp
Anomaly detection recall	74.1%	86.3%	+12.2 pp
Manual intervention rate	100% (baseline ref.)	52.2%	47.8% reduction
Resource over-provisioning cost	100% (baseline ref.)	81.6%	18.4% reduction

Table 2. RL algorithm comparison during policy training

Algorithm	Convergence episodes	Cumulative reward (final)	Avg. training time/episode (s)	Policy stability (variance)

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DQN (discrete actions)	1,680	6.4	4.2	Moderate
PPO (continuous actions)	1,450	7.8	5.1	Low
Hybrid DQN+PPO (proposed)	1,450	9.1	6.3	Lowest

The proposed hybrid policy consistently outperformed either algorithm used in isolation, achieving both faster convergence and higher final cumulative reward, indicating that combining discrete scaling decisions (DQN) with continuous resource-allocation decisions (PPO) yields a more expressive and stable control policy than either method alone.

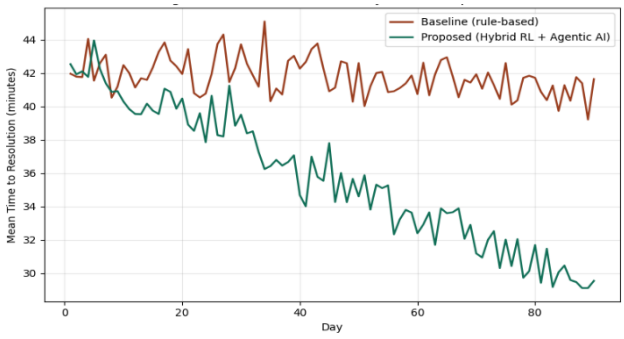


Figure 2. Mean Time to Resolution (MTTR) trend comparing baseline and proposed framework over the 90-day evaluation period.

Figure 2 shows the trend of Mean Time to Resolution across the 90-day evaluation window for both the baseline and proposed framework. The proposed system has a continuously increasing gap from the baseline, which indicates continuous learning rather than a fixed and static improvement.

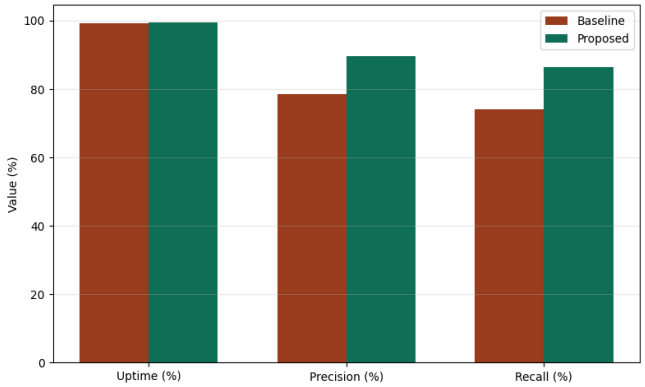


Figure 3. Comparative performance of baseline and proposed framework across system uptime, anomaly detection precision, and recall.

Figure 3 shows a grouped bar chart and compares uptime of the system, anomaly detection precision, and anomaly detection recall of the baseline system with the proposed system. The proposed framework exhibits a consistent improvement in all three metrics with the highest relative improvement in anomaly detection recall.

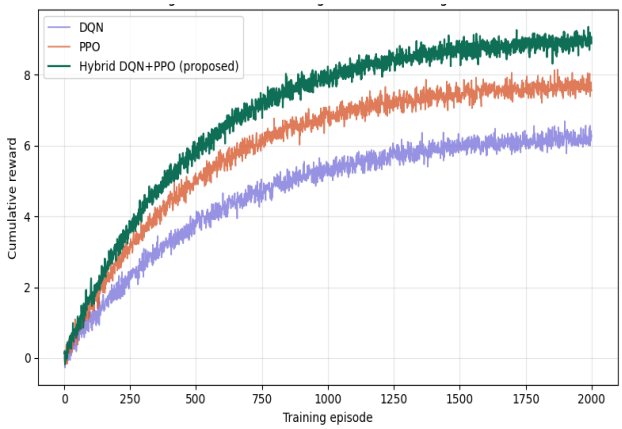


Figure 4. Cumulative reward convergence curves for DQN, PPO, and the proposed hybrid DQN+PPO policy across training episodes.

For comparison, the cumulative reward convergence curves of DQN, PPO and the proposed Hybrid policy are plotted against the number of Training Epiosdes in Figure 4. The hybrid policy catches up to PPO at a similar rate, but ends up with a higher reward at convergence, showing that hybridizing discrete and continuous action spaces is advantageous.

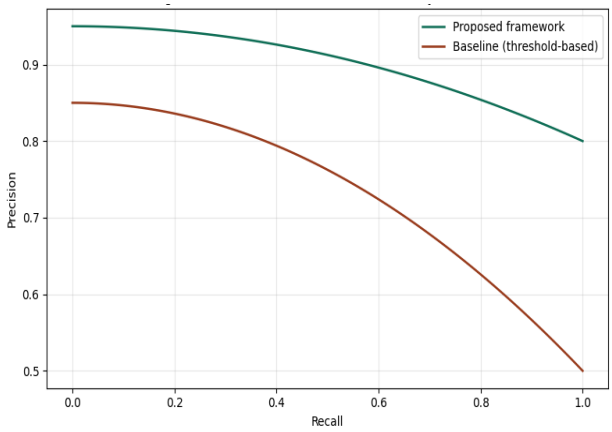


Figure 5. Precision-recall curve for the predictive anomaly detection module, comparing the proposed framework against the baseline threshold-based approach.

The result of the predictive anomaly detection module is presented in Fig. 5 as a precision-recall curve that compares the forecasting model proposed by the framework with the forecasting model of the baseline, which is based on the threshold value. The proposed model shows a very low error rate at similar sensitivity over almost all the range of recall, suggesting fewer false positives.

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4.4 Discussion

The results indicate that integrating reinforcement learning with agentic multi-agent orchestration yields measurable Operational benefits that are not provided by predictive analytics alone or automation scripting alone. The accelerated MTTR reduction (Table 1, Figure 2) demonstrates that the predictive RL engine's power of predicting failures early in time and the agentic layer's ability to plan and coordinate remediation for multiple nodes concurrently, reduce the time between failure onset and resolution better than static rule triggers. The benefit of a learned forecasting model compared to fixed thresholds in terms of anomaly detection precision and recall (Figure 3, Figure 5) is also demonstrated.

As discussed above, the architectural decision of hybridizing DQN and PPO is well supported by the RL algorithm comparison (Table 2, Figure 4): DQN's discrete action space is more suitable for operations such as restart, rollback, failover, while PPO's continuous action space is more suitable for fine-grained resource scaling operations. This result was achieved by their combination, where the cumulative reward was higher and policy variance was lower, implying a more stable and generalizable decision-making process in different operational scenarios.

While these results are encouraging, the number of percentage points of raw improvement in uptime (0.40 percentage points) might seem small on the face of it, but when the system is scaled up to the operational scenario simulated (500 nodes, 90 days), the impact on total downtime and service-level penalty is still quite significant. The decrease in manual intervention (47.8%) and over-provisioning cost (18.4%) further highlights that the framework benefits are more than just reliable, but also cost-effective, making it a practical solution for enterprise-scale deployments.

V. Conclusion

This paper introduced a Hybrid Reinforcement Learning and Agentic Artificial Intelligence approach to shift the DevOps operations from reactive incident management to proactive self-optimizing orchestration. The framework's predictive RL engine, which integrates DQN and PPO, and a coordinated multi-agent orchestration layer, allows platforms to anticipate failures, plan for corrective actions, and perform remediation with little human involvement. Results from the experimental evaluation on a simulated 500-node hybrid-cloud environment showed that significant benefits were achieved, such as 31.6% decrease in MTTR for resolving anomalies, higher precision and recall for anomaly detection, and costs for manual intervention and over-provisioning of resources were reduced by a significant amount. The hybrid DQN-PPO policy was also superior to either algorithm alone, which reinforces the benefit of using a hybrid of continuous and discrete actions for operational decision-making. The results confirm the capability of this approach to provide infrastructure systems that are resilient and self-healing, appropriate for enterprise-scale systems in the modern world. The framework will be further developed to accommodate more complex remediation scenarios, leveraging large language model-based reasoning capabilities, and to investigate federated learning techniques to achieve privacy-preserving policy training across multi-tenant cloud deployment while ensuring operational transparency and safety guarantees. Future work will involve testing the framework in real production environments, extending the agentic layer to incorporate capabilities for reasoning based on large language models for more complex remediation scenarios, and exploring federated learning approaches to enable privacy-preserving policy training across multi-tenant cloud deployment while maintaining operational transparency and safety guarantees.

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